

Welfare measurements with heterogeneous agents.

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Motivation

THE MISSING INDEX

- The canonical infinite-horizon framework with heterogeneous consumers, standard in macroeconomics and finance, lacks a preference-based index that consistently quantifies the welfare impact of economic policies.
- The classic money-metric indices, equivalent and compensating variation, fail two consistency requirements in this setting: they are non-additive across sets of policies, and their predictions can depend on the assumed status quo or on the order in which alternatives are implemented.
- The paper delivers a positive result on when these failures become quantitatively negligible.

Money-metric welfare indices

DEFINITIONS

For consumer i with lifetime utility V_i , wealth w_i and a policy moving the environment from ϑ^0 to ϑ^1 , the equivalent and compensating variations solve

$$V_i(w_i + EV_i; \vartheta^0) = V_i(w_i; \vartheta^1), \quad V_i(w_i; \vartheta^0) = V_i(w_i - CV_i; \vartheta^1).$$

- Additivity would require the index of a bundle of policies to equal the sum of the indices of its elements; exact additivity fails for general preferences, and with it order-independence.
- The question is under what restrictions the money metric becomes approximately additive, so that policy rankings aggregate.

Main results

EXISTENCE, UNIQUENESS AND THE ADDITIVE LIMIT

- **Well-posedness.** For any pair of policies and discount factor $\beta < 1$, the equivalent variation exists and is unique (Proposition 1).
- **Additive limit (Theorem 1).** As consumers become patient, aggregate equivalent variation converges to a surplus representation,

$$\lim_{\beta \rightarrow 1} EV_{p,p'} = \frac{S(\Delta q^0, \Delta Y)}{\bar{v}_x},$$

which is additive on the set of policies and measurable with respect to aggregate income. The surplus consists of a terms-of-trade component and an aggregate-income component.

- Near-additivity for β close to one delivers consistent rankings for transient policies.

Application: stimulus variants in the Polish economy

QUANTIFYING THE APPROXIMATION ERROR

- Preference and income distributions are extracted from Polish micro-econometric data; welfare effects of variants of a stimulus package are computed exactly and via the additive approximation.
- For policies affecting the economy within the first four quarters the approximation error is no greater than 1.5% of the total welfare effect; within twelve quarters, no greater than 5%.
- As a byproduct the paper contributes to the aggregation literature: exact aggregation of money-metric welfare classically requires Gorman polar form; the patient-consumer limit provides an alternative route for transient policies.

| Implications

FOR APPLIED POLICY EVALUATION

- Welfare statements in heterogeneous-agent models gain a consistent money-metric foundation for the empirically dominant case of short-lived policies evaluated by patient consumers.
- Policy bundles can be evaluated component-wise and summed, and rankings are invariant to the order of implementation up to the quantified approximation error.
- The result connects the applied macro practice of reporting consumption-equivalent welfare numbers to an explicit preference foundation.

Thank you.

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